Assignment 9: Sponsored Search Market (VCG and GSP)

Batch B · Group 14

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Market (Pricing-1, slide 57). Slots a, b, c have click-through rates 10, 4, 0. Advertisers x, y, z earn 7, 6, 1 per click. Find the efficient allocation and the VCG prices, then analyse the GSP auction (slide 54): truthful bidding, equilibria, revenue.

1. Efficient allocation

Value of a slot to an advertiser = CTR × value per click.

	x	y	z
a (10)	70	60	10
b (4)	28	24	4
c (0)	0	0	0

The best assignment is x→a, y→b, z→c, worth 70 + 24 + 0 = **94**. The next best is y→a, x→b, worth 88.

2. VCG

Each winner pays the harm it causes the others: their best total value without it, minus their total value with it (slide 56).

	Others' value without it	Others' value with it	Price	Per click	Payoff
x	y→a 60 + z→b 4 = 64	y→b 24 + z→c 0 = 24	**40**	4	30
y	x→a 70 + z→b 4 = 74	x→a 70 + z→c 0 = 70	**4**	1	20
z	70 + 24 = 94	70 + 24 = 94	**0**	0	0

VCG revenue is **44**. A winner's price depends only on the others' reports, so reporting the true value is optimal. (The same method on the slide 55 market gives 13 and 3, as on slide 56.)

3. GSP

Truthful bids (7, 6, 1). x takes a and pays y's bid: 6 × 10 = 60, payoff 10. y takes b and pays 1 × 4 = 4, payoff 20. z takes c and pays 0. Revenue is 64.

Truthful bidding is not an equilibrium. If x bids 5, it drops to slot b and pays z's bid: 1 × 4 = 4. Its payoff rises from 10 to 28 − 4 = **24** (figure below).

Equilibria. Both profiles below survive every unilateral deviation.

Bids (x, y, z)	Allocation	Payoffs (x, y, z)	Revenue	Welfare	Best deviations
(5, 4, 2)	x, y, z	30, 16, 0	48	94	x bids 3: 28 − 8 = 20. y bids above 5: 60 − 50 = 10. z takes b: 4 − 16 = −12.
(3, 5, 1)	y, x, z	24, 30, 0	34	88	y bids 2: 24 − 4 = 20. x bids above 5: 70 − 50 = 20. z takes b: 4 − 12 = −8.

In the second equilibrium y (value 6) holds slot a while x (value 7) holds b, so 6 units of welfare are lost. z bids freely because slot c has zero clicks.

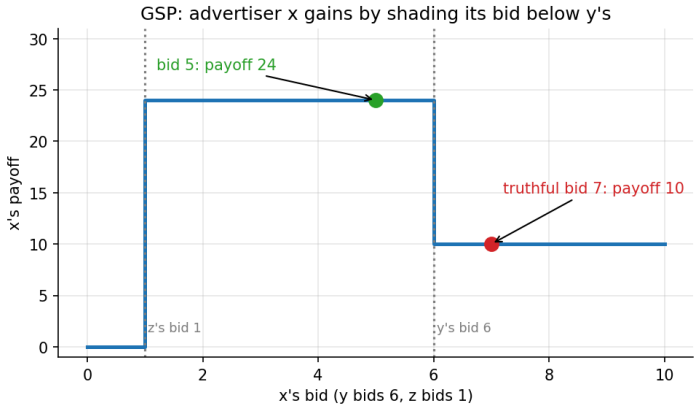


Figure. Payoff to x as its bid varies, with y bidding 6 and z bidding 1.

4. Comparison

Mechanism	Revenue	Welfare
VCG	44	94
GSP, truthful (not an equilibrium)	64	94
GSP equilibrium (5, 4, 2)	48	94
GSP equilibrium (3, 5, 1)	34	88

- VCG is truthful and efficient, with revenue 44.
- GSP is not truthful and has several equilibria, some inefficient.
- GSP revenue can be above (48) or below (34) VCG, depending on the equilibrium.